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Devops

Devops => Devlopment + Oprations

"Product ko banane se lekar live aur Maintain karne tak ka Process => Devops Hai.

Git
Jenkins
Docker
AWS

Git => Git ek Version Control System Hai jo Code files ke Changes ka record rakhta Hai aur Purane Version per wapas Jaana main Help Karta Hai.

Git -> Local => Computer main Code manage Karta Hai.
Github -> Global / online => Internet par Repository Store / share Karta Hai.

Jenkins => Jenkins ek automation tool Hai jo Software ke Build, Test aur Deploy Jaise kaam automatically karne mai help karta Hai.

Jenkins Pipeline : ek automated Process Hai jisme Software ke Build, Test aur Deploy Jais Steps ko ek Sequence (Step by Step) main arrange Kiya Jata Hai.
CI / CD pipline ko Automate karne wala tool

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CI => Continuous Intergration
CD => Continuous Deployment

Docker => Docker ek tool Hai jo Application aur uske required environment ko Container main Package karke easily Run karne main help karta Hai.

AWS => (Amazon web Services) ek Cloud Platform Hai Jo internet ke Through Server, Storage database aur other IT Services provide karta Hai.

Server => Network Per connected user ya devices ka data, Applications ya Services Provide karta Hai.

Client => Service maangta Hai.
Server => Service deta Hai.

Network => Ek Medium of Communication Hai

Networking => Client + Network + Server / Service.
Mobile => Client
Internet => Network
Youtube => Server
Video => Service

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URL = Uniform Resource locate

Https ://www.Raystec.com / Courses / devops
↓ ↓
Protocol Domain Name

Domain Name : Domain Name website ka human Name Hota Hai. Jo ip address ki jagah use kiya jata Hai

https :// 127.0.0.1 :8080 / Courses / devops
↓ ↓ ↓
Protocol IP Address Port Number

IP Address => Hotel ka Naam / Address
Port Number => Hotel ka Room Number

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Devops (github)

Sabse pehle Github Par New Repository Banai. Us Repository Me Jakar <Code> Se URL Copy Kiya.

Desktop ek New folder banaya
folder Ko open Kar ke Address bar Mai CMD Type Kiya
↓
git init ⇒ Enter
↳ git clone Copy URL Past https:// Enter

Deskop per jo folder Hai Uske Ander wale folder Ko open Kar ke "Project" Ko Past Karna Hai.

Next :- Address bar Par CMD Karna Hai

CMD :-

1. ↳ git add . → Enter
2. ↳ git Commit -am "done" ⇒ Enter
   ↳ git Push

GitHub :- to Local
git pull

Ek Project Par Ek Se jyda team Kaam Kar Rahi Hai to Alag alag Branches Banana padti Hai.

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Github Par Repo

↓

Repo URL Copy

↓

Computer Par folder

↓

CMD open folder mai

↓

git init

↓

git clone URL

↓

Repo folder me CMD

↓

git Add .

↓

git Commit -m "done"

↓

git Push

↓

git branch Rakesh (Rakesh branch dal)

↓

git Push origin Rakesh

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Devops (Servers)

Servers :- Provid Service

1. Apache Tomcat => only Java ke liye or Choti Applications ke liye.

2. wildfly (J.boss) -> Java ke liye or Badi Applications ke liye

3. Nginx -> Front hand ke liye use hota hai.

1. Apach Tomcat Run karne ke liye
   - Apach Tomcat Root folder
   - Go to The Bin
   - Duble click on Startup (windows batch file)

- Stop karne ke liye
  - Apach Tomcat Root folder
  - Go to The Bin
  - Duble Click on shutdown (windows batch file)
  - Alternative : (Ctrl + C)

Steps to Change Apache Tomcat Port Numbers
- Apach Tomcat Root Folder
- Go to The Conf directry
- Open Server.xml in a text editor (Notepad)
- Find (Ctrl + F) Connector port line.
- Change the Port numbers,

before port = "8080: After = "8081"

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2. Run wildfly (Jboss) IpAdd : localHost:8080

- Wildfly Root Folder
- Go to the bin directory
- Double-click on standalone (windows batch file)

- Stop wildfly (Jboss)
  - Press Ctrl + C in Comand Prompt
  - Press Y when Prompted

Step to chang wildfly (Jboss) - Port number

- wildfly Root Folder
- Go to The Standalone directory.
- Navigate to The Configiweation directory.
- Open Standalone.xml in text editor (Notpad)
- Find the Socket-binding line.
- Change the Port number
  before : 8080 After 8081

3. Run nginx → ip Add : localHost

- nginx Root folder
- CMD - Address Bar
- Type Start nginx

Close ke liye → nginx -S stop
nginx ka Difalt port no. 80 hota hai
Step to chang Port number Nginx

- Root Folder → ~~nginx~~
- Conf
- nginx.Conf open in Notepad
- Find The listen 80;

koi change na Ho to
nginx -reload Calanah

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Server Par Html File kese Run kare.

Apach Tomcat

- Chat GPT se html file banwai.
- Go to webapps in Tomcat Root Folder
- Go to Root Folder
- Past html file
- Chack File name extension
  View → Show → File Name extensions
- Remove .txt
- After → Run in (localhost:8080/ index.html)

  ↓

  To file ka name diya h.

Wildfly

- Wildfly Root folder
- Go to welcome - Contect
- Past html file
- Chack File name extension
- Remove .txt
- Run in (localhost:8080/ file ka name)

Nginx

- nginx Root folder
- Go to html folder
- Past html file
- Chack file name extension
- Remove .txt
- Run in (localhost:/ file Name)

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(Docker)

Docker Virtual Containar

Docker ek open Source Software platform hai. Jiska istemal applications ko develop. Ship aur Run. ke liye kiya jata hai.

Docker aapke Code ko ek chote lightweight packed mein Pack kar deta hai. Jise Container kehate hai.

Docker par bahut Saare Software ki ready-made Images available hoti hai - Jaise Java, Python, Tomcat, wildfly. nginx, MySql

Image = Kisi Software ka - Docker me calane ka Ready Package.

Docker Image banane ke liye Hum Dockerfile Banate hai. Dockerfile Docker ko Batati hai ki Image kaise banani hai.

Docker file ka extensions nahi hota hai

Desktop Folder -> Dokerdemo
Uske ander file => index.html

Docker Demo k ander -> Docker file. isme .txt nahi Hona Chahiye.

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File me likho

From tomcat : 10

Copy index.html /usr/local/tomcat/webapps/
Root/index.html

Docker demo folder me CMD open karo
Command
docker build -t mytomcat .

- Docker build => Docker image banao
- -t mytomcat => image ka naam mytomcat Rakho
- . → isi Current folder ki files use karo

Command
docker run -d -p 8080:8080 --name Mytomcat mytomcat

Nginx
From => nginx
Copy → index.html /usr/share/nginx/html/
index.html

Wildfly
From => quay.io/wildfly/wildfly
Copy => Root.war /opt/Jboss/wildfly/
- Standalone/deployments/Root.war

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Run karne ke Commands

- docker build -t myapp -t myapp Name dai
- docker run -d -p 8080:80 myapp -> nginx / apache
- docker run -d -p 8080:8080 myapp -> tomcat / wildfly